Diffusion-limited adsorption to a spherical geometry: The impact of curvature and competitive time scales

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Although the time scale governing diffusion-limited transport of soluble species from solution onto a planar interface is well understood, the time scale governing transport onto a spherical interface is not. The time scales that have been proposed in the literature for spherical interfaces do not capture the correct asymptotic behavior for increasing bubble radius and do not capture previously reported experimental observations of the effect of concentration. This paper develops a diffusion-limited time scale that is dependent on an intrinsic length scale termed the spherical depletion depth. The time scale is determined by considering a specific example of diffusion-limited transport of surfactant species to a water-air interface and is verified using numerical simulations and experiments. This newly derived diffusion time scale will have a significant impact on our understanding of fundamental phenomena at spherical fluid-fluid and fluid-solid interfaces, especially those involving micrometer and nanometer length scales.

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I. INTRODUCTION

The increasing interest in droplets and bubbles in microfluidic devices has driven us to reconsider the problem of dynamic transport to spherical interfaces. The competing time scales involved in coupled diffusion and ad/desorption transport to highly curved interfaces complicates the modeling and control of numerous physical and chemical processes including: the stabilization of colloidal suspensions and emulsions [1], the synthesis and growth of nanoparticles in solution [2], particle dissolution [3], and reactions that occur on catalytic particles and surfaces [4]. Defining the time scales and length scales pertinent to the underlying transport equations allows for appropriate modeling of specific situations and development of techniques to extract the relevant transport parameters. The goal of this paper is to define the appropriate time scale for diffusion controlled transport of surfactant to a curved fluid-fluid interface. Although this work is presented in terms of a specific example, surfactant transport, the time scales and approach are robust and can be applied to other diffusion-reaction systems.

Molecules and particles diffuse via Brownian motion along concentration gradients. In many situations such gradients are induced by the removal of material from the bulk to populate a surface. An example is the adsorption of surfactants to liquid or solid interfaces [5,6]. A characteristic time scale for transport via diffusion can be written as,

$$\tau_D = \frac{l^2}{D}, \tag{1}$$

where l is a characteristic length scale across which diffusion must occur and D is the diffusion coefficient. In an unbounded geometry where there is no obvious geometric length scale, a natural length scale is obtained by considering the mass removed to an interface compared with the mass available in a region of the bulk near that interface. Equating the number of molecules of a species accumulated at equilibrium on the interface with the number of molecules of that species available in a volume element of the bulk solution leads to a length scale called the depletion depth, which characterizes the width of the mass transfer boundary layer, or the steepness of the concentration gradient. For example, for surfactant adsorption to a planar interface, Fig. 1(a), the number of molecules adsorbed to a unit area of the interface, dA, at equilibrium is given by $\Gamma_{eq}dA$, where Γ_{eq} is the equilibrium surface concentration. The number of molecules in a volume element adjacent to the interface is given by $C_{bulk}h_p dA$, where h_p is the depletion depth and C_{bulk} is the surfactant concentration in the bulk. Thus, the depletion depth for a planar interface is given by $h_p = \Gamma_{eq}/C_{bulk}$ and the characteristic time scale for diffusion to a planar interface using Eq. (1) is given by,

$$\tau_{Dp} = \frac{h_p^2}{D}. \tag{2}$$

Since the equilibrium surface concentration is dependent on the bulk concentration through an isotherm, the planar depletion depth h_p is also an intrinsic length scale for the surfactant that depends on bulk concentration and molecular properties of the surfactant.

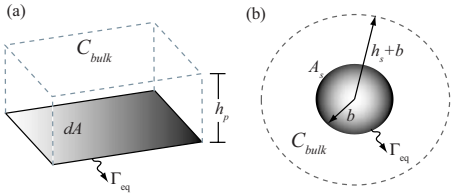


FIG. 1. (Color online) Schematic representation of the depletion depth for (a) a planar interface, h_p and (b) a spherical interface, h_s.

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The characteristic length scale for diffusion to a spherical interface is more complicated than that for a planar interface, since the radius introduces an additional length scale with which to scale the problem. Jin *et al.* [7] recently showed that by scaling the boundary condition for diffusion to a spherical interface, assuming the radius b as the relevant length scale and in the limit $b/h_p \ll 1$, one obtains a characteristic time scale for diffusion given by,

$$\tau_{Dj} = \frac{h_p b}{D} \tag{3}$$

suggesting that diffusion is faster for increasing curvature or smaller radius [7]. The radial dependence of the diffusion-limited time scale is due to the increase in the ratio of solution volume to surface area with decreasing radius. In other words, at any given time a smaller radius has a larger number of surfactant molecules adjacent to the interface than a larger interface. Thus a smaller radius reaches equilibrium faster. Recognizing this radius dependent diffusion time scale represented a change in thinking for the field.

Motivated by the analysis of Jin *et al.*, we interpreted experimental data and trends in the context of Eq. (3). However outside of the limit that $b/h_p \ll 1$, this scaling cannot capture two observations found in the literature. The first is that the time scale does not approach that of the planar interface as the radius grows large and the curvature approaches zero, as we would expect. Instead, Eq. (3) suggests that the time scale continues to grow without bound as the radius increases. The second arises when diffusion competes with other mass transport processes, leading to competing time scales. For example, Jin *et al.* [7] consider the competition between diffusion, which drives surfactant to the interface, and kinetic barriers, which govern the rates at which molecules near the interface adsorb and desorb. Comparing the diffusion time scale given in Eq. (3) with a kinetic time scale, τ_k, for ad/desorption at an interface yields a dimensionless ratio analogous to the Damkoehler number used in reaction-diffusion systems [8]. For the case considered by Jin *et al.*, $b/h_p \ll 1$, the ratio is independent of concentration. A concentration independent ratio results because in the limit of $b/h_p \ll 1$, $C_\infty \ll \Gamma_{eq}/b$ and therefore Eq. (3) is confined to a region of low concentrations. A lack of dependence of the dimensionless ratio of time scales on concentration suggests that concentration does not determine the conditions at which the dominant mass transport mechanism shifts from diffusive to kinetic. However, experimental observations described by Pan *et al.* [9] for short chain alcohols and the semianalytical study performed by Lin *et al.* [10] both show that kinetic processes become increasingly important at high concentration. These observations suggest that a more appropriate time scale for a spherical geometry should result in a limiting behavior for large radii and a ratio of kinetic to diffusion time scales that is dependent on concentration for intermediate to large values of b/h_p.

In the present paper, we further develop the scaling analysis presented by Jin *et al.* for diffusion-limited transport to a spherical interface (solid or free) for all values of b/h_p and verify the scaling using both numerical simulations and two distinct experimental data sets using a nonionic surfactant as a test species. This new time scale is then compared with the time scale for kinetics to determine the governing transport mechanism at all radii and concentration conditions. The framework developed here will yield a quantitative means of determining the importance of surfactant kinetic exchange at an interface, i.e., when kinetics are slow enough to alter the overall rates of transport to interfaces.

II. DIFFUSION-LIMITED TIME SCALE FOR A SPHERICAL GEOMETRY

To control and understand processes involving mass transport, we need to formulate the correct scaling behavior. This is particularly true when two or more processes compete. Here, we develop a new time scale for the general problem of diffusion mass exchange at spherical interfaces which overcomes the two limitations described earlier. We validate this scaling using numerical simulations solving for diffusion-limited mass transport of surfactant to a spherical interface. The result is an improved scaling, which will provide a tool to better design and characterize dynamic surface tension results as well as surfactant-induced phenomena at interfaces, such as tipstreaming [11], detergency [12], and emulsion and film stability [1,13].

Similar to the planar depletion depth, a spherical depletion depth is derived by considering a spherical interface in an unbounded geometry. As shown schematically in Fig. 1(b), the number of molecules occupying the surface at the equilibrium surface concentration, Γ_{eq}, is given by $4\pi b^2 \Gamma_{eq}$. The number of molecules occupying the volume extending a radial distance h_s from the surface of the sphere is given by $\frac{4}{3}\pi C_{bulk}[(h_s+b)^3 - b^3]$. Combining these two expressions and solving for h_s leads to the expression,

$$\frac{h_s}{h_p} = \left(\frac{b}{h_p}\right)\left[\left(\frac{3h_p}{b}+1\right)^{1/3} - 1\right]. \tag{4}$$

The spherical depletion depth depends on both the intrinsic (planar) depletion depth for the surfactant, and the radius of curvature of the interface. In the limit that the radius of curvature is large compared with the intrinsic depletion depth $b \gg h_p$, the spherical depletion depth approaches the planar depletion depth $h_s \to h_p$. In the limit of small radii, $b \ll h_p$, the spherical depletion depth scales with both the intrinsic depletion depth, h_p, and the radius of curvature, b, such that,

$$h_s \cong (3b^2 h_p)^{1/3}. \tag{5}$$

The behavior of the spherical depletion depth scaled with the intrinsic depletion depth, h_s/h_p, as a function of the dimensionless length scale b/h_p is shown in Fig. 2. It is notable that the ratio h_s/h_p depends solely on the ratio b/h_p, which contains all of the relevant surfactant conditions including concentration, bubble radius, and surfactant molecular parameters.

III. NUMERICAL METHOD

To determine whether the relationship given in Eq. (4) is relevant to the diffusion-limited mass transport to a spherical interface, we use numerical simulations to determine the ap-

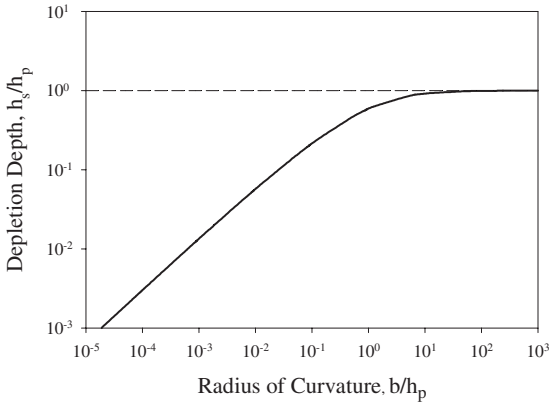


FIG. 2. Spherical depletion depth scaled by the planar depletion depth as a function of b/h_p (solid line). Dashed line represents the planar depletion depth.

propriate characteristic time scale. The bulk diffusion of surfactant is modeled using the unsteady form of Fick's law of diffusion in spherical coordinates, $\partial C/\partial t = D\partial/\partial r(r\partial C/\partial r)$, subject to a flux condition at the interface, $\partial\Gamma/\partial t = D\partial C/\partial r|_{r\to b}$ [14]. Far from the interface, a Dirichlet boundary condition applies, $C_{r\to\infty}=C_{bulk}$, and we use the initial condition $\Gamma(t=0,r=b)=0$, which corresponds to a clean interface. The flux condition is constrained by the Langmuir adsorption isotherm,

$$\frac{\Gamma_{eq}}{\Gamma_\infty} = \frac{C_{bulk}}{C_{bulk}+a}, \tag{6}$$

where a is the ratio of kinetic constants for adsorption, β, and desorption, α [15]. In this study, the diffusion coefficient is taken to be constant regardless of the proximity of the molecule to the interface. Recent studies on colloidal particles suggest that the diffusion coefficient is not constant as the particle approaches the interface [16,17]. Although not accounted for here, this type of behavior would introduce new length scales and time scales into the problem.

We solve the governing equations using a spectral analysis method to obtain the complete time-dependent concentration profile for diffusion-limited dynamics [18]. A spectral analysis was performed using Chebyshev grid points and an implicit Euler scheme for discretization in time. Convergence was verified by changing the number of grid points and decreasing the time step until a change in these parameters yielded a change in dynamic surface tension less than a prescribed tolerance, $\Delta\gamma \leq 10^{-6}$. The spectral algorithm was also verified by comparing the converged solution with the semi-analytical solution to the differential equations described by Mysels [19] and solved by the same methodology as Miller and Kretschmar [20].

IV. DIFFUSION-LIMITED TIME SCALE

Given the general form of the scaling in Eq. (1), we expect that the time scale for diffusion to a spherical interface will contain some function of the length scales h_s, h_p, and b. In the case of a planar interface, the diffusion time scale has the form of Eq. (2). However, for a spherical interface, more than one choice of scaling exists. For example, the scaling given in Eq. (3) by Jin *et al.* [7] arises from a scaling of the flux boundary condition using the radius as the characteristic length scale, $r_c=b$. If the flux boundary condition is instead scaled using the spherical depletion depth, $r_c=h_s$, *the* characteristic time scale $\tau_1=h_sh_p/D$ arises. Independently scaling Fick's diffusion equation, using $r_c=h_s$, yields a characteristic time scale $\tau_2=h_s^2/D$. Therefore, the bulk process and the surface flux are governed by different but comparable time scales, τ_1 and τ_2, whereas in the case of the planar geometry only one unique time scale results from the same scaling analysis. Here, we choose a single time scale that is a simple geometric mean of the two time scales τ_1 and τ_2. A similar approach to developing a governing scaling from multiple natural time scales is used in other systems. For example, competitive time scales are seen in the relaxation of an entangled mesh of wormlike micelles, which is governed by both a polymeric reptation time and a breaking and reforming time; in this case a simple geometric mean of the two natural time scales provides a governing time scale that is verified by experiments [21]. In the case of diffusion to a spherical interface, then, we consider the combined time scale given by the geometric mean of τ_1 and τ_2 above, such that,

$$\tau_{D,s} = (\tau_1\tau_2)^{1/2} = \frac{(h_s^3h_p)^{1/2}}{D}. \tag{7}$$

As we previously described, when $b\to\infty$ we see that $h_s\to h_p$ and thus the diffusion time scale reduces to $\tau_{D,s}\to h_p^2/D$. Therefore, as expected, the time scale given in Eq. (7) captures the asymptotic limit corresponding to a planar interface. In the limit of small radii, $b\ll h_p$, the time scale given by Eq. (7) approaches the time scale found by Jin *et al.* [7] Combining Eq. (5) and Eq. (7), the limiting time scale approaches $\tau_{D,s}\to h_pb/D$ [cf. Eq. (3)]. Therefore, the results presented in Jin *et al.* can be interpreted as an asymptotic limit of the spherical diffusion time scale presented here.

The effective lengthscale over which diffusion occurs for a spherical interface [i.e., l from Eq. (7)], is given by $l=(h_s^3h_p)^{1/4}$. This equation can be rearranged to obtain $l=(h_s/h_p)^{3/4}h_p$. Using the relationship presented in Eq. (4), l can be written as a function of b and b/h_p only. In Fig. 5 of Jin *et al.* the following scenario is presented: $b=10$ μm, $C_\infty=0.8$ μM and $h_p/b=210$, leading to a value of $h_s/h_p=0.0361$, so $l=1.74\times10^{-4}$ m and $l/b=17.4$. Even though the spherical depletion depth is much smaller than the planar depletion depth, the effective depletion lengthscale is larger than the bubble radius. In other words, using a small radius does not ensure that the diffusion lengthscale is small since the lengthscale is also dependent on concentration.

It is important to note that this newly derived time scale is independent of the isotherm selected when scaled by the planar time scale. Scaling the spherical diffusion time scale given in Eq. (7) by the planar time scale, Eq. (2), leads to

$$\frac{\tau_{Ds}}{\tau_{Dp}} = \left(\frac{h_s}{h_p}\right)^{3/2} = \left(\frac{b}{h_p}\right)^{3/2}\left[\left(\frac{3h_p}{b}+1\right)^{1/3}-1\right]^{3/2}, \tag{8}$$

which depends only on the dimensionless quantity b/h_p, [cf. Eq. (4)]. Although the intrinsic depletion depth, h_p, is determined by the isotherm, varying this quantity only serves to change the relative magnitude of b/h_p. In other words, the isotherm determines where the system conditions fall on the scaling curve, but does not change the shape of the curve. Therefore, any system undergoing diffusion-limited transport to a spherical interface is expected to follow this scaling regardless of the thermodynamic condition that governs its equilibrium.

V. NUMERICAL VALIDATION OF TIME SCALE

To validate the scaling given in Eq. (7), numerically computed dynamic surface tension curves are analyzed. Using the simulated surface concentration profiles, $\Gamma(t)$, we determine the time, τ_{num}, required to reach a given fraction, ϕ, of the equilibrium surface concentration, $\phi=\Gamma(\tau_{num})\Gamma_{eq}=$const. The dependence of τ_{num} on b/h_p is then used to test the scaling of the characteristic time for diffusion to within a leading constant. A range of values of b/h_p were obtained by (i) considering a fixed radius while varying concentration and (ii) considering a fixed concentration while varying radius. For a planar interface, the time scale obtained for all surface concentration fractions considered, $0.1 \leq \phi \leq 0.9$, agreed well with the expected scaling given by Eq. (2), verifying the technique.

Figure 3(a) compares the time scale resulting from the full diffusion-limited numerical simulations with three different relevant scalings. Note that the numerical time scale has been empirically shifted by a leading constant such that the planar region asymptotically approaches unity for direct comparison with the theoretical scaling. For experimental results, the shift factor is observed to depend on ϕ and the isotherm chosen. In the case of the numerical simulations, the correct isotherm governing the dynamics were known and therefore the shift factor only depends on ϕ. In addition, for all $\phi < 0.90$ the horizontal axis was shifted by dividing the value of b/h_p used in the numerical simulations by ϕ. This scaling is appropriate since only a fraction of the equilibrium coverage is reached at finite values of τ_{num}. Therefore, h_p should be evaluated at the instantaneous surface concentration $\Gamma(\tau_{num})$ and not the equilibrium value Γ_{eq}. In other words, the extracted time scale corresponds to a depletion depth of $h_{p,\tau_{num}}=\Gamma(\tau_{num})/C_{bulk}$ and thus

$$\frac{b}{h_{p,\tau_{num}}} = \frac{bC_{bulk}}{\Gamma(\tau_{num})} = \frac{b}{h_p\phi} \tag{9}$$

so that the time scale corresponds to the fractional coverage and not the equilibrium coverage. Figure 3 shows that τ_{num} approaches the planar time scale at large radii and the time scale given by Jin *et al.* [7] at small radii, as expected from the asymptotic analysis given in Eq. (5). The time scale given in Eq. (7) captures both this limiting behavior as well as the intermediate scaling for $b/h_p \sim O(1)$. Therefore, we conclude that a geometric mean of the two comparable time scales for bulk and surface processes does provide the correct characteristic time scale for diffusion-limited transport to a spherical interface.

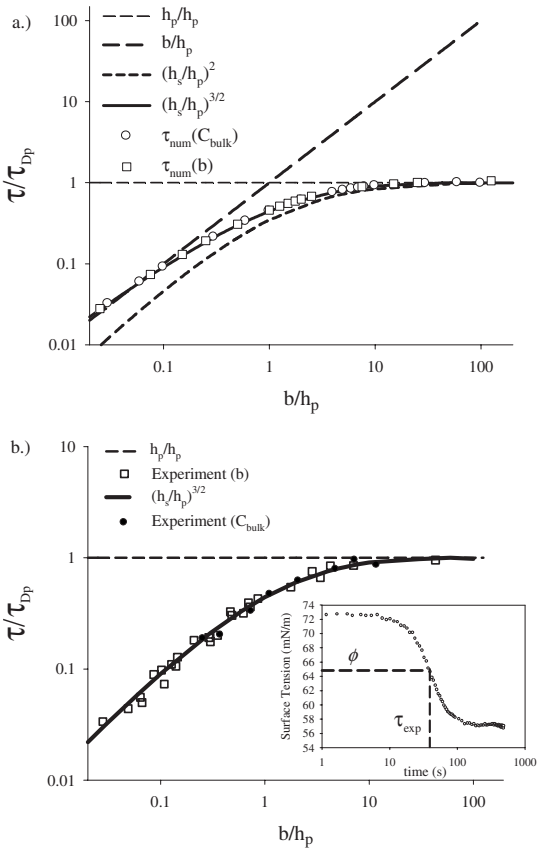


FIG. 3. (a) Comparison of three possible time scales scaled by the planar time scale as a function of b/h_p. The time scale obtained from numerical simulations is represented by open symbols. The open circles correspond to simulations conducted for changing concentration at fixed radius. The open squares correspond to simulations conducted for changing radius at fixed concentration. Note that the scaling behavior for the characteristic time is the same for both cases. The horizontal line corresponds to the time scale for a planar interface. (b) Comparison of experimental data scaled by the planar time scale as a function of b/h_p. The circles correspond to experiments where the concentration was varied at fixed radius and the squares correspond to experiments where the bubble radius was varied at fixed concentration.

VI. EXPERIMENTAL VALIDATION OF TIME SCALE

Experimental validation of this time scale was performed by measuring the surfactant dynamics for varying bubble radii and concentration, and conducting a similar analysis to that described in the numerical validation section. Since the time scale is independent of the isotherm governing the equilibrium condition at the interface, any surfactant system can be used to examine the scaling. Conventionally, surfactant dynamics are tested experimentally using shape analysis of a pendant drop of fixed bubble radius and varying bulk concentrations [22]. Results from the literature were used to test our scaling. To test the scaling with varying bubble size at fixed surfactant concentrations, we have built a device in which a bubble is formed at the end of a glass capillary. Capillaries of different radii were used. The surface tension was inferred from a time varying measurement of the pressure jump across the interface and radius of the bubble, the

TABLE I. Parameters for the surfactants used in this study along with the isotherm employed in the analysis.

Surfactant	Γ_∞ [μ mol/m^2]	a [mol/m^3]	κ	n	β [m^3/mol/s]	D [10^{10} m^2/s]	R_{DK} [μm]	b [μm]	Isotherm
Butanol [32]	6.7	47	0	0	1.6[a]	8.8	80	~500	L
Hexanol [32]	6.0	3.7	0	0	35[a]	7.0	3.4	~500	L
Nonanol [33]	6.4	3.0	−4.97	0.29	100	6.3	1.0	~800	GF
Decanol [34]	7.0	0.97	−5.03	0.26	10	6.0	8.4	~800	GF
Model	6.0	8.0×10^{-6}	0	0	100	7.0			L
$C_{12}E_8$ [26]	5.3	2.3×10^{-6}	13.2	0.50	5.6[b]	4.0	14	17–800	GF

[a]These values were obtained from Joos and Serrien [31].
[b]Calculated from scaling limits as explained in text.

details of which can be found elsewhere [23]. The surfactant used for this study is a nonionic alkyl polyglycolether surfactant, $C_{12}E_8$, for which dynamic surface tension measurements at fixed radius have been reported in the literature [24].

The analysis of the experimental data is performed via the same method described in the numerical validation section, except that the characteristic time scale is extracted from experimental dynamic surface tension curves [Fig. 3(b) inset] [23] instead of the dynamic surface concentration curves, which are directly accessible in simulations. Three concentrations were selected, C_{bulk}=0.0006, 0.0025, and 0.006 mol/m^3, which are all well below the expected critical micelle concentration of 0.1 mol/m^3 [24]. The bubble sizes used for these experiments were b=22, 29, 56, 92, 155, and 1500 μm. The selection of different concentrations extended our access to a wider range of b/h_p, due to the limited range of experimentally accessible bubble radii. The experimental time scales were extracted from the data by choosing $0.1<\phi<0.9$, which is converted to the corresponding surface tension using the equation of state for $C_{12}E_8$,[24–26] and extracting the experimental time required to obtain this fractional coverage (i.e., surface tension) for a given concentration. An example is shown in the inset of Fig. 3(b). To determine the dependence of the time scale on concentration, data extracted from Fig. 1 of Lin *et al.* [24] was also used. The same technique of extracting an experimental time scale was used. Again a constant fractional surface coverage was chosen, $0.1<\phi<0.9$, and the governing equation of state was used to calculate the surface tension corresponding to each concentration for the selected fractional coverage. Specifically for $C_{12}E_8$, the Generalized Frumkin isotherm was used to calculate the intrinsic lengthscale, h_p, and thus, the planar diffusion time scale. The parameters for the isotherm were taken from the literature [25] and verified experimentally by a nonlinear fit to equilibrium surface tension data, see Table I. Although the isotherm used does not change the scaling, it is important that the isotherm used fits the equilibrium surface tension data and that the same isotherm is used to scale the entire data set.

The results of the experimental analysis for both varying concentration and varying radius are shown in Fig. 3(b). The scaled data clearly follows the scaling given in Eq. (7). Since the diffusion-limited scaling is observed for the entire range of experimental conditions considered for this surfactant, we conclude that the dynamic surface tension response must be diffusion limited. This is in contrast to many models that have been proposed for this surfactant [24,25,27,28], in which kinetic exchange at the interface is introduced to describe the observed dynamics. In fact, Fig. 3(b) suggests that the correct choice of isotherm along with diffusion-limited dynamics should describe the dynamics without having to introduce additional kinetic barriers at the interface.

These two distinct experimental data sets have further validated the correct time scale for diffusion-limited surfactant transport to a spherical interface. In the next section, we address some of the implications of the results shown in Fig. 3(b) on our understanding of surfactant dynamics and on the importance of surfactant kinetics at an interface.

VII. COMPETING TIME SCALES

Now that the diffusion time scale to a spherical interface has been verified, we can determine the relative impact of competing time scales on transport to interfaces. If the dynamics of a surfactant exhibit a time scale that differs from the form of the diffusion-limited time scale, then it is possible that kinetic exchange at the interface or another process (e.g., convection [29]) is competing with diffusion to the interface. We can determine the relative importance of kinetics and diffusion via the ratio of the characteristic time scales for the processes [30], often referred to as the Damkoehler number in the reaction-diffusion literature. To illustrate this point, consider a surfactant that obeys Langmuirian kinetics [20,30], where the evolution of surface concentration at a static interface is given by

$$\frac{d\Gamma}{dt} = \beta C_{bulk}\Gamma_\infty\left(1-\frac{\Gamma}{\Gamma_\infty}\right) - \alpha\Gamma, \tag{10}$$

where β is the adsorption rate constant and α is the desorption rate constant. At equilibrium Eq. (10) reduces to the Langmuir isotherm, [cf. Eq. (6)]. The characteristic kinetic time scale is obtained from the solution to the differential equation in Eq. (10), and is given by

$$\tau_k = \frac{1}{\beta C_{bulk}+\alpha}. \tag{11}$$

Therefore, the ratio of the two time scales is given by

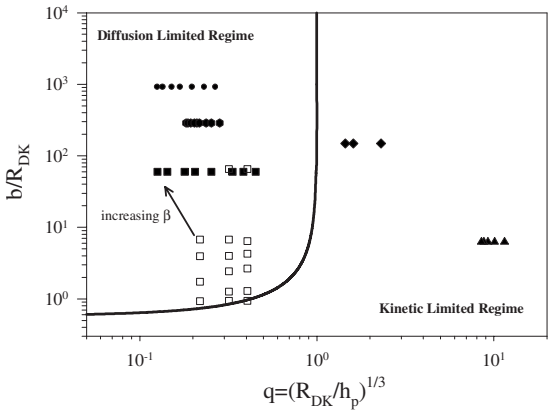


FIG. 4. The effect of bubble radius and concentration on the governing transport of a soluble adsorbing species following Langmuirian kinetics. The line corresponds to equality of the diffusion time scale and the kinetic time scale. Diffusion-limited dynamics exist far to the left of the line and kinetic-limited dynamics exist far to the right of the line. The points correspond to data extracted from literature (filled symbols) or conducted with varying bubble size (open symbols). ▲ butanol; ◆ hexanol; ⬭, nonanol; ● decanol; ■ $C_{12}E_8(C_{bulk})$; and □ $C_{12}E_8(b)$

$$\Lambda = \frac{\tau_{Ds}}{\tau_k} = \frac{(h_s^3 h_p)^{1/2}}{D}(\beta C_{bulk} + \alpha). \tag{12}$$

A similar analysis was performed by Pan *et al.* who used the planar diffusion time scale to explain the observation that surfactants exhibit kinetic-limited dynamics at high concentrations [9]. Jin *et al.* used the analysis of Pan *et al.* and the scaling given in Eq. (3) to demonstrate how curvature affects the relative magnitudes of the two time scales [7]. In particular, Jin *et al.* noted an intrinsic length scale, $R_{DK}=D/(\Gamma_\infty b)$, at which the two time scales are equal ($\Lambda=1$) and a transition from diffusion-limited to kinetic-limited transport is expected. Using the corrected scaling for a spherical interface, Eq. (12), we determined the critical radius, R_{crit}, at which the kinetic and diffusion time scales are equal by solving

$$\left(R_{crit}\left[\left(\frac{3h_p}{R_{crit}}+1\right)^{1/3}-1\right]\right)^3 h_p\left(\frac{\beta C_{bulk}+\alpha}{D}\right)^2 = 1. \tag{13}$$

Solving Eq. (13) for R_{crit} and scaling by R_{DK}, the intrinsic surfactant length scale derived by Jin *et al.*, the new critical radius is given by

$$\frac{R_{crit}}{R_{DK}} = \frac{2}{\sqrt{12-3q^2}-3q}, \tag{14}$$

where $q=(R_{DK}/h_p)^{1/3}$ is a dimensionless ratio of two intrinsic length scales for the surfactant. Thus, the critical radius now depends on the bulk concentration as well as material properties as experiments and analysis have previously shown [9,10,30–32]. Figure 4 shows the dependence of the scaled R_{crit} on the value of q. The solid line corresponds to the radius at which the two time scales (kinetic and diffusion) are equal. Values far to the left of the solid line correspond to diffusion-limited dynamics and values to the far right correspond to kinetic-limited dynamics. Note that as $q\to 1$, $R_{crit}/R_{DK}\to\infty$. Figure 4 is consistent with the expectation that as bubble radius decreases the dynamics become more kinetic limited. In addition, it captures the experimentally and theoretically observed dependence of the controlling mechanism on surfactant concentration [9,10,30–32]. As concentration increases, q increases and therefore the system moves toward the kinetic-limited regime. The scaling analysis put forth by Jin *et al.* was independent of concentration and therefore inconsistent with these experimental investigations. However, by using the correct diffusion time scale, the scaling analysis is in agreement with both the radius dependence presented by Jin *et al.* and the concentration dependence observed experimentally and discussed in Pan *et al.* It should be noted that Fig. 4 is general for systems that undergo a Langmuirian kinetic exchange at the interface. A similar analysis could be performed for systems governed by other kinetic models. Such analysis is useful in determining the parameter space in which a transition from diffusion limited to kinetic-limited transport can be expected.

Among other surfactants, short chain alcohols exhibit kinetic-limited behavior in experiments since high concentrations are needed to achieve appreciable changes in surface tension [9,31,32]. We have plotted b/R_{DK} for four short chain alcohols in Fig. 4, namely butanol, hexanol, nonanol, and decanol, using parameters and dynamic curves extracted from the literature, listed in Table I. The placement of the data on Fig. 4 suggests that butanol is the most likely of the surfactants considered to exhibit kinetic-limited behavior. As the tail length increases, the short chain alcohols become more diffusion limited (i.e., move further to the left of the equality line). From experiments reported in the literature, it is observed that butanol is completely kinetic limited, while hexanol exhibits mixed dynamics [31]. Nonanol [33] and decanol [34] exhibit little or no observable kinetics. Fainerman and Miller later argued that short chain alcohols are completely governed by diffusion-limited dynamics at the concentrations tested [32]. However, based on the scaling arguments presented here, butanol and hexanol will not exhibit purely diffusion-limited dynamics. To better understand whether short chain alcohols exhibit diffusion-limited, mixed, or kinetic-limited dynamics we return to the spherical depletion depth scaling arguments.

Extracting experimental time scales from published data [32] and plotting in a similar way to the data shown in Fig. 3, we can determine whether kinetics or diffusion control the dynamics. To compare both the kinetic and diffusion-limited time scales on the same plot, we scale the Langmuir kinetic time scale by the planar diffusion time scale, which leads to

$$\frac{\tau_k}{\tau_{Dp}} = \frac{R_{DK}}{h_p} = q^3. \tag{15}$$

It is notable that this ratio of time scales is given by the same ratio of intrinsic surfactant length scales that defines the transition from diffusion to kinetic-limited transport [cf Eq. (14)] Represented on a log-log plot as a function of b/h_p, Eq. (15) yields different functional forms depending on the type of experiment performed. For fixed bubble size and varying concentration, Eq. (15) is a line with slope unity and intercept $I=\log(R_{DK}/b)$. For fixed concentration and varying

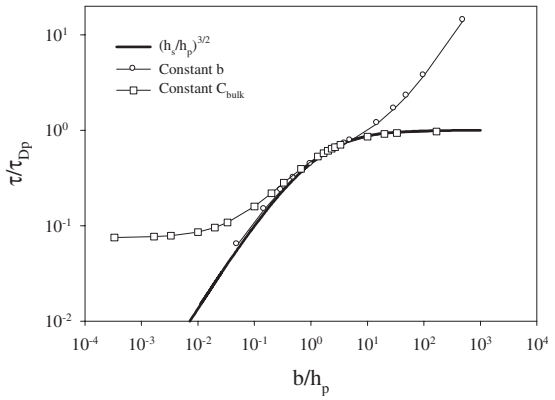


FIG. 5. Characteristic time scale obtained from mixed kinetic-diffusion simulations (see Table I Model parameters) scaled by the planar diffusion time scale and plotted as a function of b/h_p. The two different experiments (fixing concentration or bubble radius) exhibit a different form of transition from kinetic-limited dynamics to diffusion-limited dynamics.

bubble radius, Eq. (15) is a constant. To test the transition from the spherical diffusion scaling to the kinetic scaling, we used the spectral method to simulate mixed diffusion-kinetic dynamics, substituting Eq. (10) for Eq. (6) and using model surfactant parameters given in Table I. The simulated time scales are shown in Fig. 5, demonstrating that the transition in controlling mechanism occurs as expected for each of the two types of experiments described.

Returning to the short-chain alcohols considered earlier, recall that the position of the points for both butanol and hexanol in Fig. 4 indicated that the dynamics should follow the kinetic-limited and not the diffusion-limited scaling. Figure 6 shows the dimensionless time scales extracted from the reported dynamic surface tension curves [32]. It is clear from this figure that butanol and hexanol follow the kinetic-limited time scale. In fact, the time scales observed in the experiments are orders of magnitude larger than the diffusion-limited time scale. A kinetic-limited transport is the

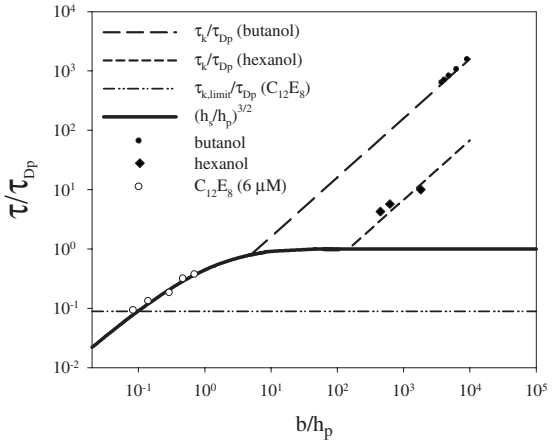


FIG. 6. Diffusion and kinetic time scales, scaled with the planar depletion depth, as a function of b/h_p for short chain alcohols and $C_{12}E_8$ at the highest concentration for different radii. Butanol and hexanol follow the kinetic time scale, while $C_{12}E_8$ follows the diffusion-limited time scale.

same result that was concluded by Joos and Serrien by fitting a simple exponential to dynamic surface tension data for both surfactants [31]. On the contrary, Fainerman and Miller concluded that butanol and hexanol show diffusion limited behavior through a short time scaling analysis. The short time scaling analysis is not sufficient to argue whether diffusion or kinetics dominate the transport dynamics. Instead the scaling analysis presented here should be followed.

A similar analysis was performed on two longer chain alcohols, nonanol and decanol, which closely follow the diffusion-limited time scale at higher concentrations. These data are not shown in Fig. 6 because deviations are observed at lower concentrations that cannot be due to kinetic or diffusion time scales and are more likely due to other phenomena such as convection [29], evaporation [32], or depletion of surfactant.

Figure 6 can also be used to provide physical limits on rate constants. Experiments performed for varying bubble radii at fixed concentration will exhibit a constant characteristic time scales as the bubble size decreases and the dynamics transition to kinetically controlled. In other words, the kinetic time scale does not depend on the bubble size. For the entire range of bubble radii that we tested, we do not observe a transition to a constant characteristic time scale (cf. open symbols in Fig. 6). Although we cannot obtain the precise value of the adsorption constant β without observing the transition to kinetic-limited dynamics, the limiting time scale corresponding to the smallest bubble radius can provide an upper limit on the kinetic time scale, and therefore, a lower bound on the adsorption rate constant. The ratio of time scales for this data point yields the inequality $\tau_k/\tau_{Dp} < 0.096$, which leads to $\beta > 5.6$ m^3 mol/s through Eq. (15). Using the isotherm parameters listed in Table I along with this minimum value of the adsorption rate constant, we plot the $C_{12}E_8$ data on Fig. 4. The scaled data appears on the diffusion-limited side of the $\Lambda = 1$ line. Since we used the limiting value of β, this data is forced to lie on the line where the two time scales are equivalent, as expected. The arrow in Fig. 5 indicates the direction that the experimental points would shift with larger values of β. Note that even though $C_{12}E_8$ follows the Generalized Frumkin isotherm, using a Langmuirian assumption for the kinetic time scale yields qualitative agreement with our experimental observations and yields insight into possible limits on the adsorption rate constant, and subsequently, the desorption rate constant.

VIII. CONCLUSIONS

In summary, we have determined the correct time scale to describe diffusion-limited transport to a spherical interface, which captures the effects of both radius and bulk concentration on the transport of species to the interface, and exhibits the appropriate asymptotic behavior in the limits of both large and small bubble size. The time scale is validated by numerical simulations and experimental data. The new scaling is used to demonstrate a method of analyzing the shift from diffusion-limited to kinetic-limited surfactant dynamics and assessing limiting values of the kinetic rate constants. For example, through a competitive time scale analysis we

were able to show that butanol exhibits kinetic-limited behavior as previously suggested, and we have determined limits on the value of the adsorption rate constant for $C_{12}E_8$. In addition, we are able to quantitatively determine the dependence of the dynamics on kinetic transport. In the case of $C_{12}E_8$, the analysis presented suggests that surface kinetics play a less important role in the transport of the surfactant to an interface than previously thought.

Although the scaling analysis presented in this paper is developed considering the case of surfactant adsorption to interfaces, this scaling applies generally to any physical problem where the flux condition at the spherical interface is solely governed by diffusion and there is no bulk convection. For example, these results could be used to explain why larger drug particles dissolve slower than smaller particles [3]. The simplicity of the scaling will allow more complex kinetic mechanisms to be considered, including the balance of diffusion and reaction in the rapidly growing field of heterogeneous catalysis at nanoparticle interfaces. As we gain control of nanoscale and microscale interfaces in a wide variety of industrial processes, the elucidation of this fundamental scaling is crucial.

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